

Using Trust for Weight Design in the Observer

We redefine a time-dependent trusted neighborhood for each agent i as :

$$\mathcal{LN}_i^{(j)}(t) = \left\{ l \in N_i \mid O_i(l, t) \geq \theta_{\min} \right\} \cup \left(\{0\} \text{ if } \mathcal{LT}_{i,j}(t) \geq \theta_{\min} \right)$$

This is the set of neighbors that legitimate agent i classifies as its legitimate neighbors at time t . Denote

$$n_{il}^{(j)}(t) = \left| \mathcal{LN}_i^{(j)}(t) \right| + 1, \forall l \in \mathcal{LN}_i^{(j)}$$

The weight matrix that priority the local estimation term is defined as

$$w_{il}^{(j)}(t) = \begin{cases} \frac{1}{n_{il}^{(j)}(t)}, & l \in \mathcal{LN}_i^{(j)}(t) \\ 0, & \text{otherwise.} \end{cases}$$

✓ Ensures resilient estimation

✓ Isolates malicious vehicles